

ROMANSY 2026

Topics on Robotics & Vision

Scent-Based Landmark and Communication for EM Hostile Environments

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16th September 2026



- An important application of interest is **disaster scenarios** [6–9].
- Temporary communication infrastructure can be hastily installed.
 - However these can't necessarily penetrate the rubble of collapsed buildings and can be generally the limiting factor of their applicability [10].

During a disaster, it is necessary for the robots to explore as quickly and efficiently as possible, and **minimising repeated coverage** of any area in the initial exploration phase helps reduce exploration time.



Figure: The practice of copying natural processes for engineering applications: The idea behind biomimicry is that nature has already solved the challenges and that engineers can use these techniques to improve their design.

- We can use **scents** which has significant research behind it [11–16] along with Molecular Communication (MC), where chemical are encoded [17].

This work proposes MC for mobile robotics for use in **cave exploration/disaster scenarios** for **rapid exploration of the environment**.

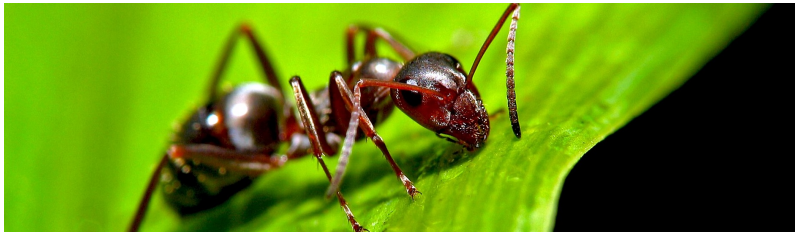


Figure: Ants create complex behaviour by using pheromones to communicate.

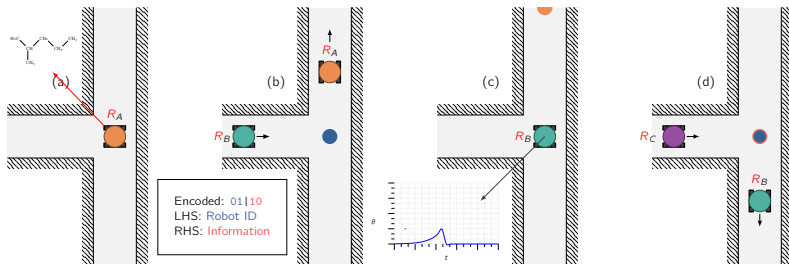


Figure: A diagram describing the scent-based land-marking.

- (a) When a robot R_A enters a cross-road, it sprays the floor with a compound which contains information about its identity and which path it took
- (b) Robot B (R_B) enters and enters the same cross-roads R_A did but detect a chemical.
- (c) (R_B) read the chemical signal and determines which way R_A went and decides to pick a different route and adds this information to the chemical pool.
- (d) (R_C) approaches.

- A PDE can have numerous solutions depending on the boundary conditions.
- A solution can be derived using the “thin-film solution” [18]:

$$c(|x| > 0, |y| > 0, |z| > 0, t_0) = 0,$$

$$c(x = 0, y = 0, z = 0, t_0) = M_0 \delta(x)\delta(y)\delta(z),$$

$$c(|x| \rightarrow \infty, |y| \rightarrow \infty, |z| \rightarrow \infty, t) = 0,$$

- M_0 is the initial mass injected into the environment, (kg)
- x is the Cartesian propagation dimension, (m)
- t_0 is the initial time, and (s)
- $\delta(x)$ is the one-dimensional Dirac delta function. (-)

- Under these conditions, the solution with an unbounded domain in x , y and a boundary at $z = 0$, can be expressed as follows:

$$C(x, y, z, t) = \frac{M}{\sqrt{(4\pi D_x D_y D_z t)^3}} \exp\left(-\frac{x^2}{4D_x t} - \frac{y^2}{4D_y t} - \frac{z^2}{4D_z t}\right)$$

- The diffusion coefficient (D), based on environment can be estimated as [19];

$$D = \frac{2}{3} \sqrt{\frac{k_B^3 T^3}{\pi^3}} \sqrt{\frac{1}{2m_A} + \frac{1}{2m_B}} \frac{4}{P(d_A + d_B)^2},$$

In simulations it is assumed the diffusion is **time-independent**.

The model of the transmission can be described in two (2) parts:

1. The absorption phase:

$$\theta_1(x_d, x_\epsilon, y_d, y_\epsilon, t) = M - \frac{M}{4} \left[\operatorname{erf} \left(\frac{x_d}{\sqrt{4Dt}} \right) + \operatorname{erf} \left(\frac{x_\epsilon}{\sqrt{4Dt}} \right) \right] \times \left[\operatorname{erf} \left(\frac{y_d}{\sqrt{4Dt}} \right) + \operatorname{erf} \left(\frac{y_\epsilon}{\sqrt{4Dt}} \right) \right] \left[\operatorname{erf} \left(\frac{z_d}{\sqrt{4Dt}} \right) \right] \quad (1)$$

2. Dispersion (removal phase) phase:

$$\theta_0(x_d, x_\epsilon, y_d, y_\epsilon, t) = \frac{M_R}{4} \left[\operatorname{erf} \left(\frac{x_d}{\sqrt{4D_x t}} \right) + \operatorname{erf} \left(\frac{x_\epsilon}{\sqrt{4D_x t}} \right) \right] \times \left[\operatorname{erf} \left(\frac{y_d}{\sqrt{4D_y t}} \right) + \operatorname{erf} \left(\frac{y_\epsilon}{\sqrt{4D_y t}} \right) \right] \operatorname{erf} \left(\frac{z_d}{\sqrt{4D_z t}} \right). \quad (2)$$

Applications of the model can be seen in detail in [20, 21].

- To model the limits on information, and understanding of the environmental noise is required.
- Based on experiments done in [21] it was determined to be an AWGN:

$$\text{AWGN} = \mathcal{N}(\mu, \sigma^2) .$$

AWGN – Fundamental statistical noise model used in communication systems to simulate the impact of **random noise** on signals. It represents random, uncorrelated noise with constant spectral density and normally distributed amplitude.

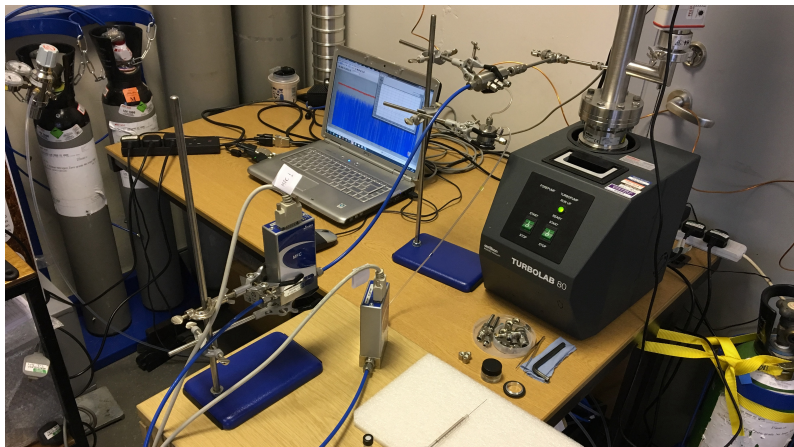


Figure: Experimental setup for the determining the environmental noise characteristic. [21]

- To embed and retain information to surface we need to use chemicals which can generate detectable vapour for a practically useful amount of time, such as *Volatile Organic Compounds* [22].
 - High vapour pressure at room temperature and common-place.
- Allow communication between animals and plants, such as attracting pollinators, or protection from predators [23–25].

Figure: *Thuja occidentalis*, where VOCs are used for communication between species.



- A robot sprays a compound onto a surface, the open surface of the liquid will enable evaporation due to the imbalance of thermodynamic equilibrium.
- Evaporation is defined not only by **geometrical parameters** such as thickness and surface area but also by **temperature** and **chemical composition**.
- However, for practical applications, simplifications can be made [21].
 - The primary propagation method here is diffusion.

Figure: *Thuja occidentalis*, where VOCs are used for communication between species.



Table: Diffusion coefficients of selected organic compounds. The diffusion values are for air [26], measured in $\text{cm}^2 \cdot \text{s}^{-1}$.

Compounds	Diffusion	Compounds	Diffusion
Cholorobenzene	0,088	Toluene	0,087
Ethylbenzene	0,075	o-Xylene	0,087

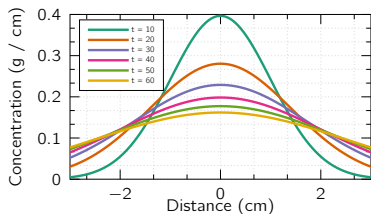
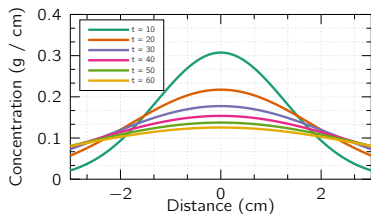


Figure: Comparison of two organic compounds concentration
 (a) Pentane ($0,0842 \text{ cm}^2 \cdot \text{s}^{-2}$) (b) n-Octyl alcohol ($0,0506 \text{ cm}^2 \cdot \text{s}^{-2}$).

To measure the chemical concentration with **high accuracy**, a chemical sensor capable of distinguishing scents is needed.

Electronic Nose

Used in detecting and **analysing odours while providing precise information**. Also known as a “fingerprint” about mixed gases and scents [27] with studies done to implement this technology to mobile robotics [28].

Mass Spectrometers

A powerful analytic technique used to **quantify known and unknown compounds** within a sample, and determine the chemical properties of different molecules, which has been use by robots as a mobile lab instead of a sensor for the robot itself [29, 30].

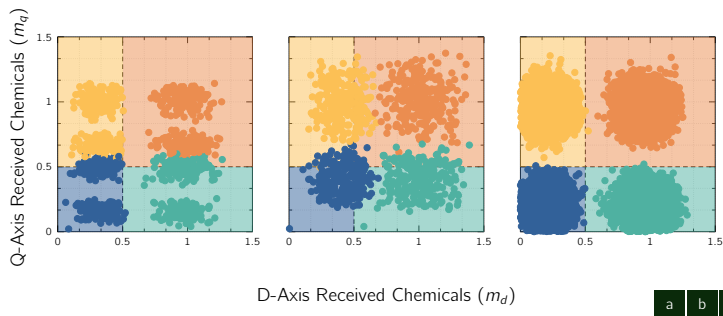


Figure: The detected concentration levels of ChQAM with a SNR of 20 dB. From left to right:

- (a) With a diffusion ratio of 3
- (b) distance of 1 cm
- (c) Listening rate of 30s.

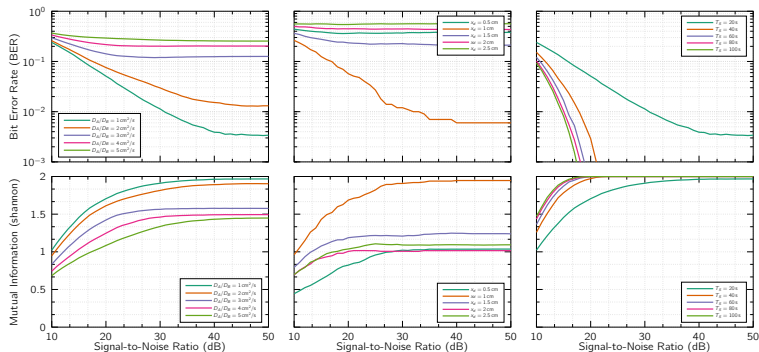


Figure: Simulation results of scent-based detection system.

From top-left first row; SNR v. BER ratio comparison of

(a) different diffusion coefficient ratios,

(b) detection distance

(c) listening duration. From left second row;

(d) signal-to-noise ratio v. mutual information of different diffusion coefficient ratios,

(e) detection distance

(f) listening duration.



- A proof-of-concept method was shown using chemicals for land-marking instead of using vision or EM waves.
- A mathematical framework was developed based on **Diffusion equation**.

Simplifications were made and a solution was derived.

- Channel calculations were conducted to see the validity of the method.
- When chemicals with **low and similar diffusivity** are chosen, it is possible for a mobile robot to detect the scent in the environment and then decode information which was left the message minutes later.
- Future-work will focus on **increasing embedded information** in chemicals using different encoding methods and conduct field experiments.

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